

Vent Simulator 2D Developer Guide

Environment

- Node version: `22`, matching `.nvmrc`.
- Install dependencies with `npm ci`.
- Run the app with `npm run dev`.

Quality Gates

Run these before release:

```
npm run type-check
npm run lint -- --quiet
npm run lint:css
npm run format:check
npm test
npm run test:e2e
npm run build
npm run check:phase-gates
```

Documentation Generation

- API docs: `npm run docs:api`.
- Codemap: `npm run docs:codemap`.
- Manual PDFs: `npm run docs:pdf`.

Architecture

The simulator separates UI components, context state, hooks, and deterministic simulation modules. The simulation layer calculates vitals, alarms, clinical assessments, visual state, waveforms, and scenario progress.

Release Checklist

- Regenerate `docs/api`.
- Regenerate `docs/CODEMAP.generated.md`.
- Regenerate `docs/manuals/\*.pdf`.
- Regenerate `artifacts/sbom.json`.
- Confirm bundle, performance, license, contrast, and raster checks.

Operational Notes

Static deployment targets are defined for Vercel, Netlify, Cloudflare Pages, and GitHub Pages. Rollback is handled by manually dispatching the CD workflow with the last known good `ref`.